
Content-Dependent Acoustic Memory

CleaRx

1 Question

Under a fixed memory budget, can Qwen3-Omni preserve more useful speech history by allocating acoustic positions according to the information they add beyond the transcript? VoxZip, the immediate prior method, pools each ASR-defined interval to its transcript-token count, fuses audio and text by addition, then prunes KV entries [1]. Its acoustic budget is therefore fixed by transcript length.

We instead allocate $b_i = g_\phi(z_i, B)$ positions to segment i according to evidence z_i , subject to $\sum_i b_i \leq B$. The budget should follow *marginal acoustic value*: evidence not already carried by text.

2 Preliminary evidence

Three frozen-Qwen3-Omni experiments support this question [3]. Mean-pooled audio reduced teacher KL from 2.136 to 1.966 in one clip and improved mean KL by 7% across ten clips, with gains on 7/10. Continuous weighting was 5.8% worse than the best fixed weight and saved no positions. Under a fixed 48-position budget, agreement-based retention improved KL by 7.96% over uniform retention, but not an audio-only allocator. Its robust contribution was the measured heterogeneity:

Benefit of four positions over transcript-only	Clean	Degraded
Exact transcript	0.0025	0.0011
Incomplete transcript	4.5444	1.7199
Conflicting transcript	4.3288	0.8755

Exact transcripts gained almost nothing; incomplete or conflicting transcripts gained much more, especially with clean audio.

3 Study

1. Allocation and representation. In multi-turn histories, measure utility at $b_i \in \{0, 1, 2, 4, 8\}$ and allocate one shared budget. At equal KV bytes, compare uniform, random, audio-only, nonlinear, and oracle allocation; separately compare mean pooling, VoxZip addition, learned summarization, and learned fusion. This isolates where memory goes from what retained positions contain.

2. Compression location. Compare the best pre-Thinker method with SpeechKV-style pooling inside early and middle Thinker layers [2]. With Qwen3-Omni frozen, measure KL, task agreement, KV memory, and latency on held-out conversations.

Adaptive allocation must beat uniform, random, and audio-only policies at equal cost; learned fusion must beat VoxZip-style addition. The result will map what acoustic evidence merits retention, how much memory it receives, and where it can first be compressed safely.

References. [1] Liu et al. [VoxZip](#). ACM MM, 2026. [2] Kim et al. [SpeechKV](#). 2026. [3] CleaRx. [Exploratory audio-memory results](#). 2026.